



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

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Course Code: DSA0401

Course Name: FUNDAMENTAL OF DATASCIENCE

1. Calculate subject averages and find the highest average

AIM:

To calculate the average marks in Math, Science, English, and History using NumPy and identify the subject with the highest average.

ALGORITHM:

1. Import the NumPy library.
2. Create a 4×4 array containing student marks.
3. Calculate the average of each subject using `np.mean()` with `axis=0`.
4. Find the highest average using `np.argmax()`.
5. Display the averages and the subject with the highest average.

CODE:

```
import numpy as np
```

```
student_scores = np.array([
```

```

[80, 75, 88, 70],

[90, 85, 82, 78],

[70, 80, 90, 85],

[85, 90, 86, 80]

])

subjects = ["Math", "Science", "English", "History"]

avg = np.mean(student_scores, axis=0)

print("Subject averages:", avg)

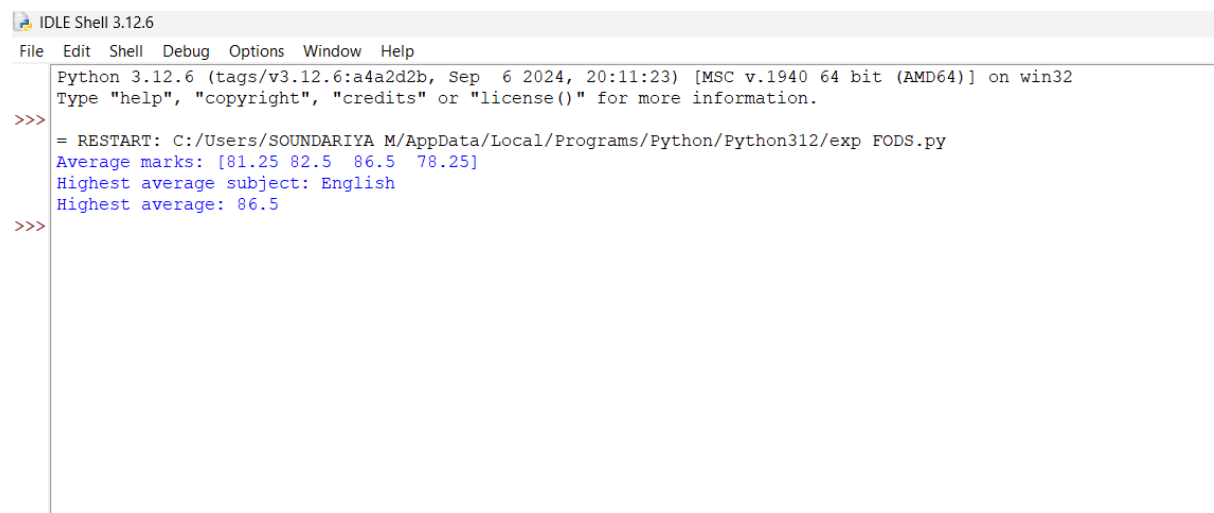
print("Highest average:", subjects[np.argmax(avg)])

```

INPUT:

A 4×4 matrix of student marks.

OUTPUT:



```

IDLE Shell 3.12.6
File Edit Shell Debug Options Window Help
Python 3.12.6 (tags/v3.12.6:a4a2d2b, Sep  6 2024, 20:11:23) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/SOUNDARIYA M/AppData/Local/Programs/Python/Python312/exp FODS.py
Average marks: [81.25 82.5  86.5  78.25]
Highest average subject: English
Highest average: 86.5
>>>

```

RESULT:

The average marks for all four subjects were calculated successfully. English has the highest average of 86.5.